ST MARY'S DSG, KLOOF

GRADE 10 **NOVEMBER 2019**

MATHEMATICS - PAPER 1

TIME: 2 HOURS **TOTAL MARKS: 137**

NAME: _____ **TEACHER:** _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This paper consists of 17 pages and 15 questions. Please check that your question paper is complete.
2. There are 4 sections (A – D). Write your name and Mathematics teacher's name on each section. Hand each section in separately.
3. Answers, where necessary, must be rounded off to two decimal digits.
4. Write neatly and show your working.
5. An approved non-programmable calculator may be used unless otherwise stated.

FORMULAE YOU MAY NEED

$A = P(1 + in)$ $A = P(1+i)^n$ $A = P(1-in)$ $A = P(1-i)^n$

$P(A) = \frac{n(A)}{n(S)}$ $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$y = mx + c$ $m = \frac{y_2 - y_1}{x_2 - x_1}$

SECTION A [36 Marks]

QUESTION 1

Simplify fully, leaving answers with positive exponents.

a) $(2y+3)(y^2-6y-8)$ (2)

$2y^3-12y^2-16y+3y^2-18y-24$

$=2y^3-9y^2-34y-24$

b) $\frac{2}{a+3}-\frac{3}{a-4}$ (4)

$=\frac{2(a-4)-3(a+3)}{(a+3)(a-4)}$

$=\frac{2a-8-3a-9}{(a+3)(a-4)} \quad =\frac{-a-17}{(a+3)(a-4)}$

c) $\frac{x^3+1}{x^2-x+1}\div\frac{2x+2}{2x}$ (4)

$=\frac{(x+1)(x^2-x+1)}{(x^2-x+1)}\times\frac{2x}{2(x+1)}$

$=x$

d) $\frac{2^{a+1}-2^{a-1}}{2^a}$ (3)

$=\frac{2^a(2-2^{-1})}{2^a}$

$=\frac{3}{2}$

e) $\dfrac{10^{2x+3}.4^{1-x}}{25^{2+x}}$ (5)

$$\frac{(2.5)^{2x+3}\cdot(2^2)^{1-x}}{(5^2)^{2+x}}$$

$$\frac{2^{2x+3}\cdot 5^{2x+3}\cdot 2^{2-2x}}{5^{4+2x}}$$

$$= 2^{2x+3+2-2x}\cdot 5^{2x+3-4-2x}$$

$$= \frac{2^5}{5}$$

f) Determine the value of $(3p+q)^2$ if $9p^2+q^2=12$ and $pq=3$. (3)

$$9p^2+6pq+q^2$$

$$= 9p^2+q^2+6pq$$

$$= 12+6(3)$$

$$= 30$$

[21]

QUESTION 2

Factorise the following completely:

a) $2p^2-4p-6$ (3)

$$= 2(p^2-2p-3)$$

$$= 2(p-3)(p+1)$$

b) $-y^4+x^4$ (3)

$$= (x^2+y^2)(x^2-y^2)$$

$$= (x^2+y^2)(x-y)(x+y)$$

c) t^2r-t^2s-r+s (3)

$$= t^2(r-s)-(r-s)$$

$$= (t^2-1)(r-s)$$

$$= (t-1)(t+1)(r-s)$$

[9]

QUESTION 3

A car costs R250 000 at the beginning of 2017. Calculate the value of the car at the end of 2019 if the depreciation is calculated at 14,5% p.a. on a reducing balance.



$A = P(1-i)^n$

$= 250\,000(1-14{,}5\%)^3$

$= R156\,256{,}59$

[3]

QUESTION 4

Given: $g(x) = 3x^2 + 2$, determine:

a) $g(-5)$ (1)

$= 3(-5)^2 + 2$

$= 77$

b) $2g(3)$ (2)

$= 2[3(3)^2 + 2]$

$= 2(29)$

$= 58$

[3]

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SECTION B **[39 Marks]**

QUESTION 5

5.1 Solve for x:

a) $\frac{x+6}{x} - x = 0$ restr: $x \neq 0$ (3)

$\frac{x+6}{x} - \frac{x^2}{x} = 0$

$-x^2 + x + 6 = 0$

$x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$

$x = 3$ or $x = -2$

b) $px - a = -qx$ (3)

$px + qx = a$

$x(p+q) = a$

$x = \frac{a}{p+q}$

c) $2^{3x+1} = 32$ (3)

$2^{3x+1} = 2^5$

$3x + 1 = 5$

$x = \frac{4}{3}$

5.2 Solve for m. Represent your answer in interval notation.

$-11 \leq 3m - 8 < 4$ (3)

$-11 + 8 \leq 3m$ $3m < 4 + 8$

$-3 \leq 3m$ $3m < 12$

$-1 \leq m$ $m < 4$ $[-1; 4)$

5.3 Solve for a and b in the following set of equations: (4)

$2a = 3 - b$ and $4b + 5a - 21 = 0$

$b = 3 - 2a$

$4(3-2a) + 5a - 21 = 0$

$12 - 8a + 5a - 21 = 0$

$12 - 21 = 3a$

$-3 = a$

$b = 3 - 2(-3)$

$= 9$

5.4 Given $q = \sqrt{c - 2ab}$, for what value/s of c will q be non-real if $a = 2$ and $b = 4$? (2)

$2(2)(4)$

$= 16$

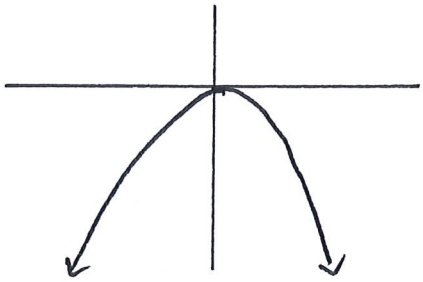
$\therefore c < 16$

[18]

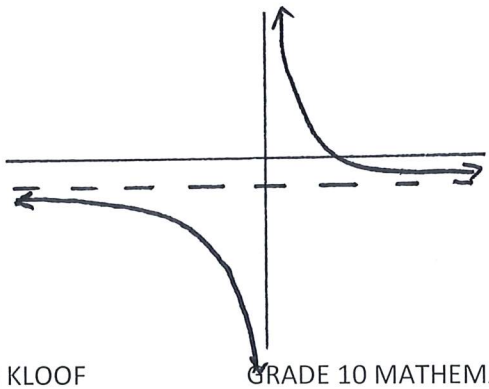
QUESTION 6

In the space below, draw rough sketches of the following graphs:

a) $y = ax^2 + q \quad if\ a < 0\ and\ q = 0$ (2)



b) $y = \frac{k}{x} + q \quad if\ k > 0\ and\ q < 0$ (2)



[4]

QUESTION 7



A family decides to buy a new speed-boat for their trip to the dam. The boat costs R 135 000. They put down a 15% deposit and pay off the balance using a Hire Purchase agreement, over 3 years. The agreement states that they will be charged 11% interest p.a.

a) Calculate the deposit amount. (1)

15% × R135000

= R20250

b) Calculate the family's monthly repayments. (4)

A = 114750 (1 + 11% . 3)

= R152617,50

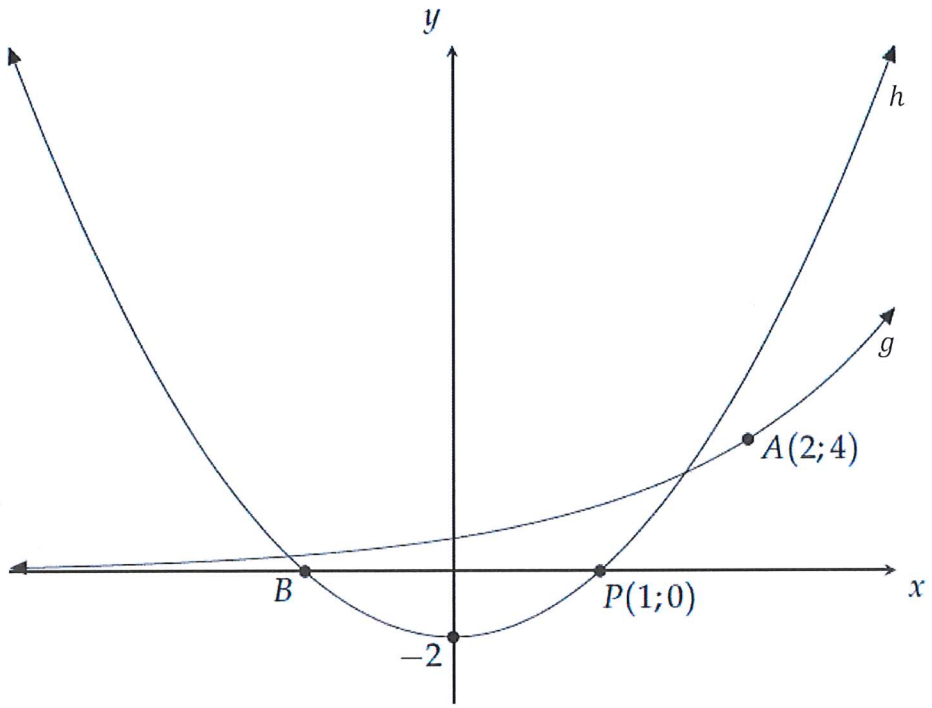
R152617,50 ÷ 36

= R4239,38 pm.

[5]

QUESTION 8

The graphs of $g(x) = b^x$ and $h(x) = ax^2 + q$ are shown below.



a) Write down the value of: (2)

i. b 2

ii. q -2

b) Show that $a = 2$ (2)

$y = ax^2 - 2$

$0 = a(1)^2 - 2$

$2 = a$

c) Write down the co-ordinates of B. (1)

$(-1;0)$

d) Write down the equation of the asymptote of $g(x)$. (1)

$y = 0$

e) Write down the equation of the axis of symmetry of $h(x)$. (1)

$x = 0$

f) For which values of x is $h(x)$ a decreasing function? (2)

$x \in (-\infty ; 0)$

g) Write down the minimum value of the function $j(x) = h(x) + 3$ (1)

min value = 1

h) Write down the range of the function $k(x)$, which is formed by reflecting $h(x)$ over the x axis. (2)

$y \in (-\infty ; 2]$

[12]

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SECTION C **[39 Marks]**

QUESTION 9

9.1 A group of hikers arrive at a campsite and decide to start arranging their sleeping bags in the following pattern. Figure 1 is made up of 4 sleeping bags.



Figure 1

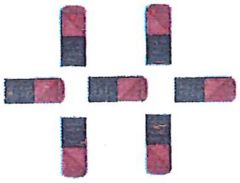


Figure 2

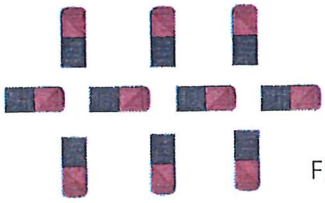


Figure 3

a) Write an expression for the number of sleeping bags in the nth figure of the pattern. (2)

$T_n = 3n + 1$

b) If they made 5 figures out of the sleeping bags, how many hikers were there? (2)

$T_5 = 3(5) + 1 = 16$

c) A few more hikers joined the group and placed their sleeping bags according to the pattern. If there were now 46 hikers in total, how many figures would be made? (2)

$46 = 3n + 1$

$45 = 3n$

$15 = n$

9.2 A snake in the campsite has shown the following pattern of growth (in cm) over the past 4 months:



2; 9; 28; 59

Write down the general formula (T_n) which could be used to calculate the snake's length at any stage over the 4 months. (3)

$T_n = 6n^2 - 11n + 7$

[9]

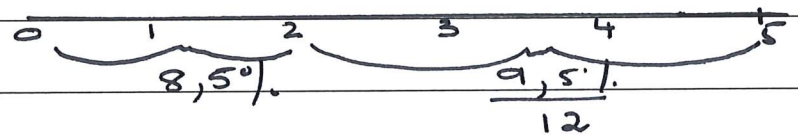
QUESTION 10



Mr Downey decided to invest R100 000 at the beginning of 2014. The interest for the first 2 years was 8,5% p.a., compounded annually. At the end of the second year, the bank changed the interest rate to 9,5% p.a, compounded monthly.

What amount will Mr Downey have in his account at the end of 2019?

100 000



$A = 10000\left(1+8{,}5\%\right)^2\left(1+\frac{9{,}5\%}{12}\right)^{4\times 12}$

$= R171886{,}42$

[5]

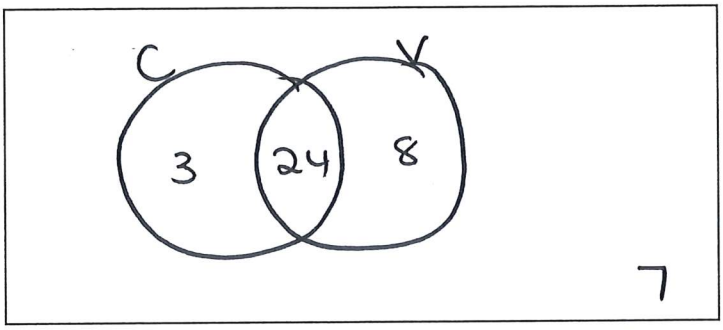
QUESTION 11



In a group of 42 Grade 10 girls:

- 27 like chocolate ice-cream
- 32 like vanilla ice-cream
- 7 do not like either

a) Draw a Venn diagram to represent the information above. (5)



$27 - x + 32 = 35$
$x = 24$

b) If a girl from the group is chosen at random, calculate the probability that she:

i. Likes chocolate ice-cream only. (1)

$\frac{3}{42}$

ii. Likes chocolate ice-cream or vanilla ice-cream. (1)

$\frac{35}{42}$

iii. Does not like chocolate ice-cream. (1)

$\frac{15}{42}$

[8]

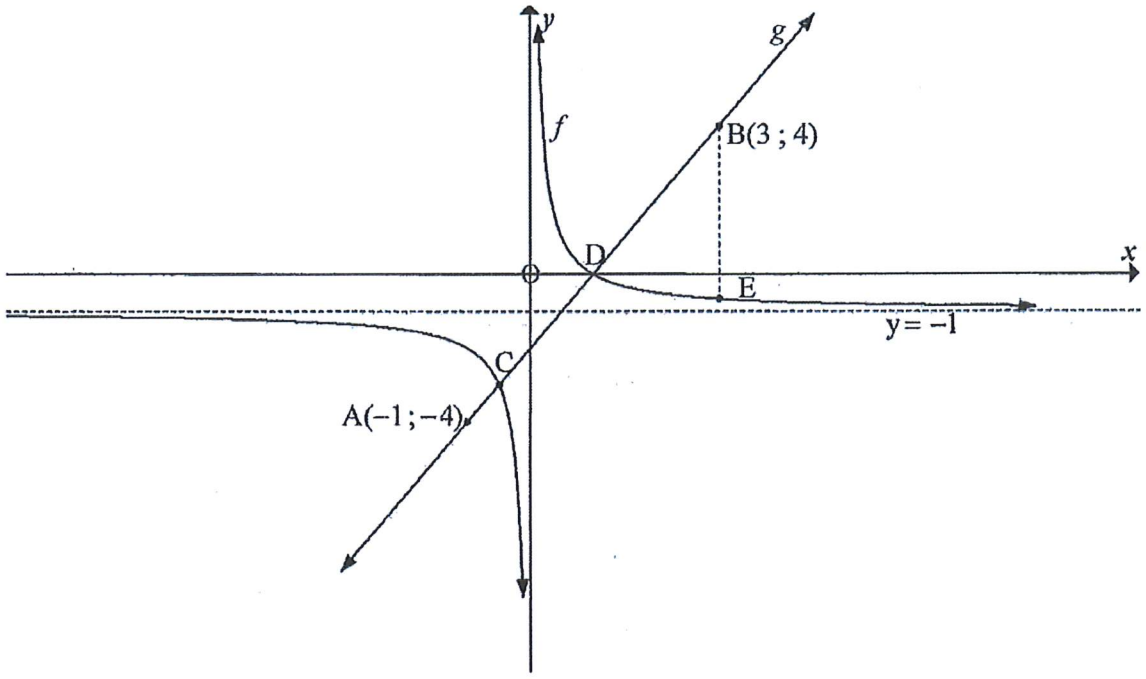
QUESTION 12

The sketch below shows the graphs of $f(x) = \frac{1}{x} - 1$ and $g(x) = mx + c$

Points A$(-1; -4)$ and B$(3; 4)$ lie on the graph g.

The 2 graphs intersect at points C and D.

Line BE is drawn parallel to the y-axis, with E on f.



a) Show that $m = 2$ and $c = -2$. (2)

$m = \frac{4-(-4)}{3-(-1)}$

$= 2$

$4 = 2(3) + c$

$-2 = c$

b) Find the equation of the line perpendicular to $g(x)$, passing through the point $(-2; 0)$. (3)

$y = -\frac{1}{2}x + c$

$0 = -\frac{1}{2}(-2) + c$

$-1 = c$

$\therefore y = -\frac{1}{2}x - 1$

c) Write down the domain of f. (2)

$x \in R$, $x \neq 0$

d) Determine the values of x for which $f(x) = g(x)$. (4)

$\frac{1}{x} - 1 = 2x - 2 \qquad x \neq 0$

$1 - x = 2x^2 - 2x$

$0 = 2x^2 - x - 1$

$0 = (2x + 1)(x - 1)$

$x = \frac{1}{2}$ or $x = 1$

e) For what values of x is $g(x) \geq f(x)$? (3)

$-\frac{1}{2} \leq x < 0$ or $x \geq 1$

f) Calculate the length of BE. (3)

$f(3) = \frac{1}{3} - 1$

$= \frac{2}{3}$

$\therefore BE = 4\frac{2}{3}$

[17]

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SECTION D **[23 Marks]**

QUESTION 13

Two events A and B are mutually exclusive.

$P(B') = 0,4$

$P(A \cup B) = 0,7$

a) Calculate $P(A)$ (2)

$P(B) = 0,6$

$\therefore P(A) = 0,1$

b) Are the events A and B complementary? Justify your answer. (3)

$P(A) + P(B)$

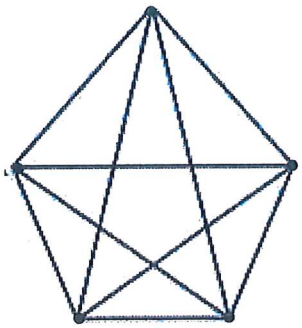
$= 0,1 + 0,6$

$= 0,7$ $\therefore$ no $P(A) + P(B) \neq 1$

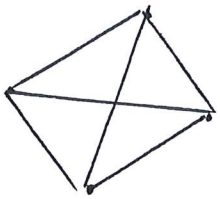
[5]

QUESTION 14

A sales person must visit 5 different shops in the Upper Highway Area and draws up a network of all possible connections between the shops, as shown in the diagram below.



a) Draw a diagram which would represent the number of possible connections between 4 different shops. (2)



1 Shop 0
2 Shop 1
3 Shop 3
4 Shop 6
5 Shop 10

b) Find a formula that you could use to work out the number of possible connections between any amount of shops. (4)

0 ; 1 ; 3 ; 6 ; 10

1 2 3 4

1 1 1

$T_n = \frac{1}{2}n^2 - \frac{1}{2}n$

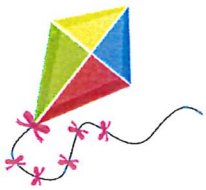
c) How many connections would be possible between 15 shops? (2)

$T_{15} = \frac{1}{2}(15)^2 - \frac{1}{2}(15)$

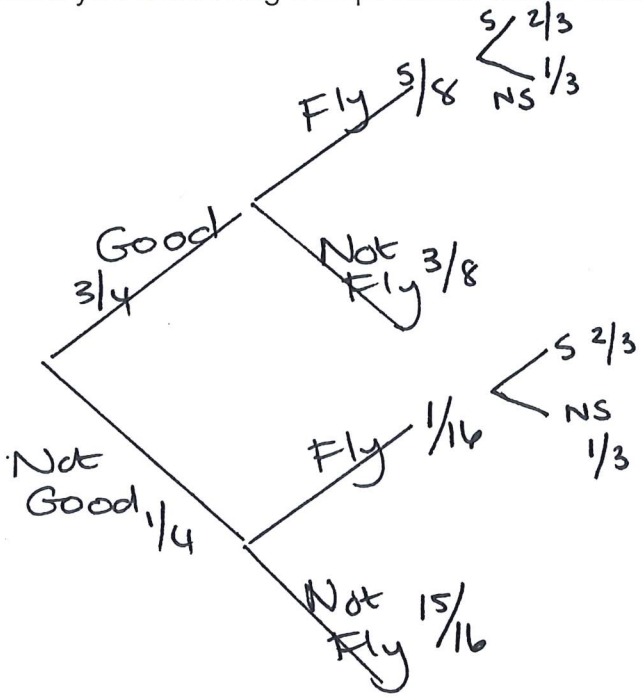
$= 105$

[8]

QUESTION 15



A little boy enjoys flying his kite. On any given day, the probability that there is good wind is $\frac{3}{4}$. If there is good wind, the probability that his kite will fly is $\frac{5}{8}$. If the wind is not good, the probability that his kite will fly is $\frac{1}{16}$. Draw a tree diagram in the space below to assist you answering the questions which follow. (6)



a) Find the probability that, whatever the wind, the kite does not fly. (2)

$\frac{3}{4} \times \frac{3}{8} + \frac{1}{4} \times \frac{15}{16}$

$= \frac{33}{64}$

b) If the kite flies, the probability that it won't get stuck in a tree is $\frac{1}{3}$. Calculate the probability that, whatever the wind, the kite will get stuck in a tree. (2)

$\frac{3}{4} \times \frac{5}{8} \times \frac{2}{3} + \frac{1}{4} \times \frac{1}{16} \times \frac{2}{3}$

$= \frac{31}{96}$

[10]